EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 5.01

C Circle Theorems, Intersecting Chords & Mensuration

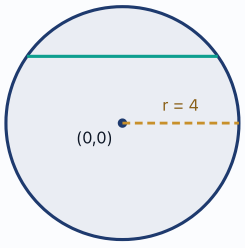
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Circle Theorems, Intersecting Chords & Mensuration Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 14x + 64$ in the domain of Circle Theorems, Similar Solids & Mensuration. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(9)$.



A 34

B 41

C 19

D 32

2 MCQ

A mathematical model in Circle Theorems, Similar Solids & Mensuration is defined by the cubic function $y_2(x) = 3x^3 - 7x^2 + 15x - 12$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

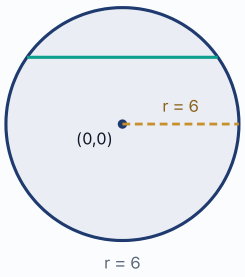
B 54

C 39

D 28

3 MCQ

In Circle Theorems, Similar Solids & Mensuration, two displacement vectors are given by $\mathbf{u}_3 = \langle 7, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 7

B 113

C 120

D 232

4 MCQ

The linear growth profile shown in the diagram for Circle Theorems, Similar Solids & Mensuration follows an arithmetic progression with first term $a_1 = 8$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 92

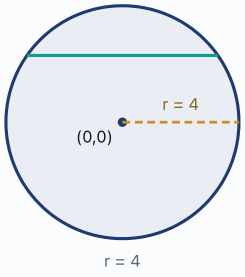
B 840

C 1380

D 750

5 MCQ

In Circle Theorems, Similar Solids & Mensuration, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 28x)\,\mathrm{d}x$.



A 104

B 128

C 132

D 53

6 MCQ

In Circle Theorems, Similar Solids & Mensuration, consider the quadratic function $f_6(x) = x^2 - 16x + 151$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(15)$.

A 231

B 223

C 136

D 216

7 MCQ

The graph illustrates the cubic model $y_{7}(x) = 3x^{3} - 8x^{2} + 33x - 19$ studied in Circle Theorems, Similar Solids & Mensuration. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 448

C 481

D 89

8 MCQ

In Circle Theorems, Similar Solids & Mensuration, two displacement vectors are given by $\mathbf{u}_{8} = \langle 8, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + \|\mathbf{u}_{8}\|^{2}$.

A 29

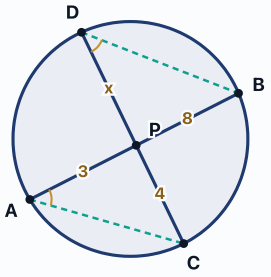
B 113

C 212

D 142

9 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 8$ cm, and $PC = 4$ cm, find the length PD.



A 5 cm

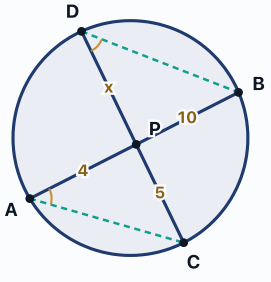
B 6 cm

C 9 cm

D 7 cm

10 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 10$ cm, and $PC = 5$ cm, find the length PD.



A 9 cm

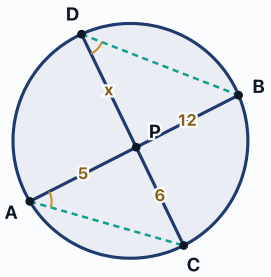
B 11 cm

C 6 cm

D 8 cm

11 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 12$ cm, and $PC = 6$ cm, find the length PD.



A 7 cm

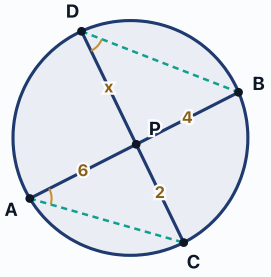
B 10 cm

C 11 cm

D 13 cm

12 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 4$ cm, and $PC = 2$ cm, find the length PD.



A 8 cm

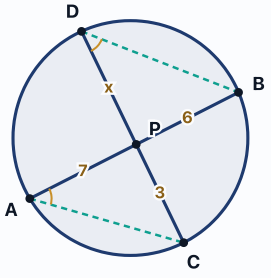
B 1 cm

C 12 cm

D 15 cm

13 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 6$ cm, and $PC = 3$ cm, find the length PD.



A 10 cm

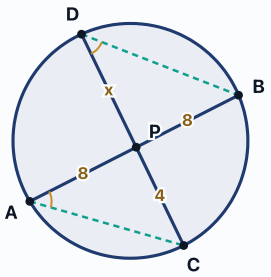
B 17 cm

C 14 cm

D 1 cm

14 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 8$ cm, $PB = 8$ cm, and $PC = 4$ cm, find the length PD.



A 1 cm

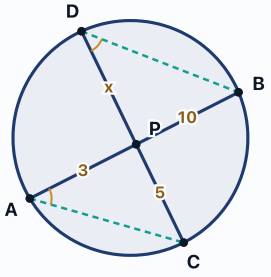
B 16 cm

C 12 cm

D 19 cm

15 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 10$ cm, and $PC = 5$ cm, find the length PD.



A 8 cm

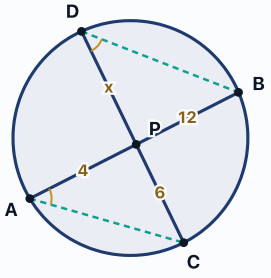
B 9 cm

C 6 cm

D 7 cm

16 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 12$ cm, and $PC = 6$ cm, find the length PD.



A 10 cm

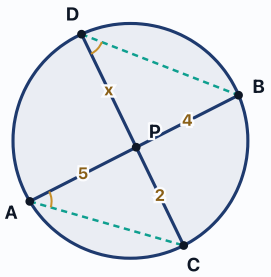
B 8 cm

C 11 cm

D 14

17 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 4$ cm, and $PC = 2$ cm, find the length PD.



A 10 cm

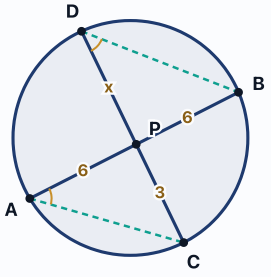
B 1 cm

C 7 cm

D 13 cm

18 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 6$ cm, and $PC = 3$ cm, find the length PD.



A 15 cm

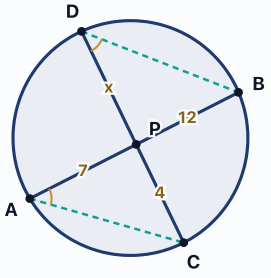
B 9 cm

C 1 cm

D 12 cm

19 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 12$ cm, and $PC = 4$ cm, find the length PD.



A 24 cm

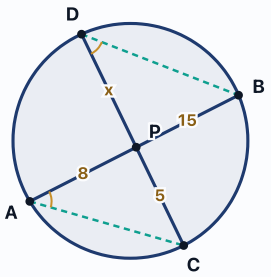
B 15 cm

C 21 cm

D 5 cm

20 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 8$ cm, $PB = 15$ cm, and $PC = 5$ cm, find the length PD.



A 24 cm

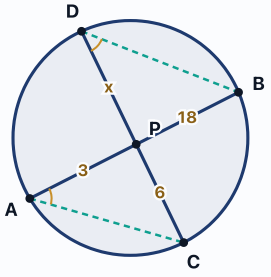
B 27 cm

C 7 cm

D 18 cm

21 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 3$ cm, $PB = 18$ cm, and $PC = 6$ cm, find the length PD.



A 9 cm

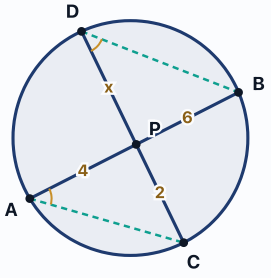
B 14

C 15 cm

D 12 cm

22 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 4$ cm, $PB = 6$ cm, and $PC = 2$ cm, find the length PD.



A 8 cm

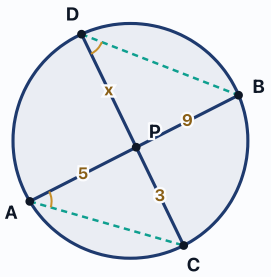
B 12 cm

C 2 cm

D 15 cm

23 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 5$ cm, $PB = 9$ cm, and $PC = 3$ cm, find the length PD.



A 4 cm

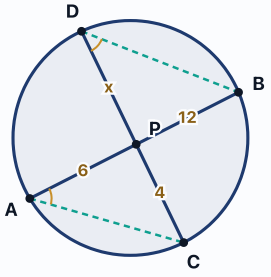
B 15 cm

C 18 cm

D 11 cm

24 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 6$ cm, $PB = 12$ cm, and $PC = 4$ cm, find the length PD.



A 18 cm

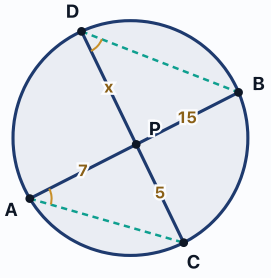
B 14 cm

C 21 cm

D 6 cm

25 MCQ

Chords AB and CD intersect internally at point P inside a circle. Given $PA = 7$ cm, $PB = 15$ cm, and $PC = 5$ cm, find the length PD.



A 24 cm

B 17 cm

C 21 cm

D 8 cm